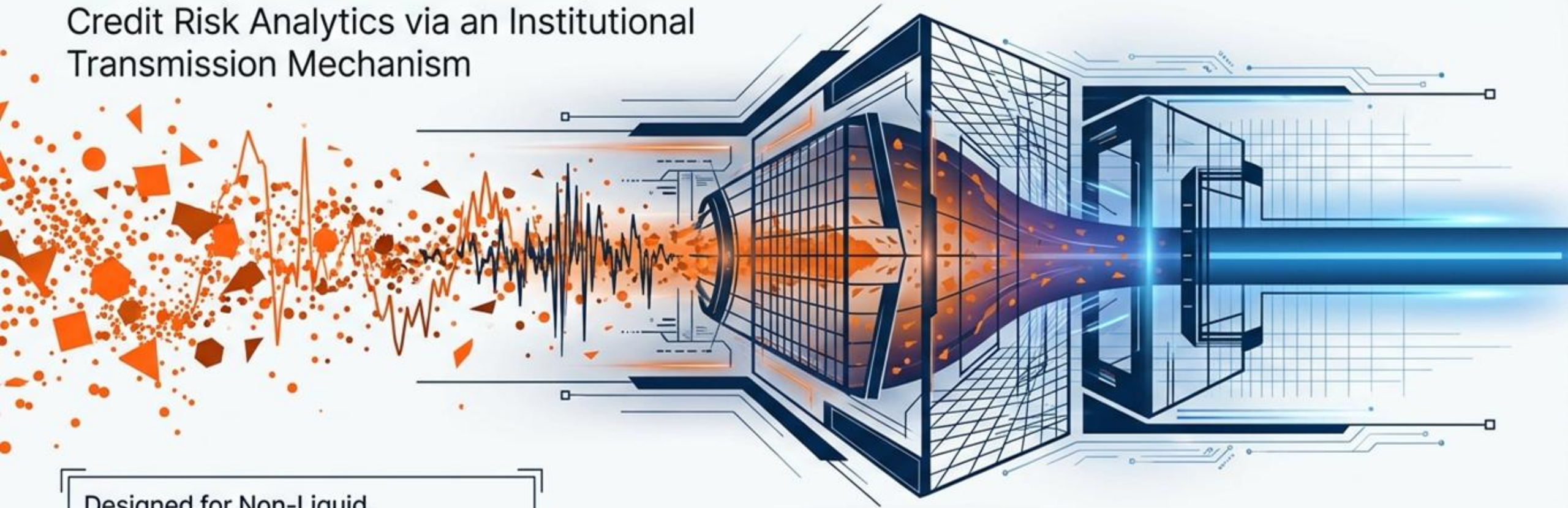


Valuation Governance in Data-Constrained Markets

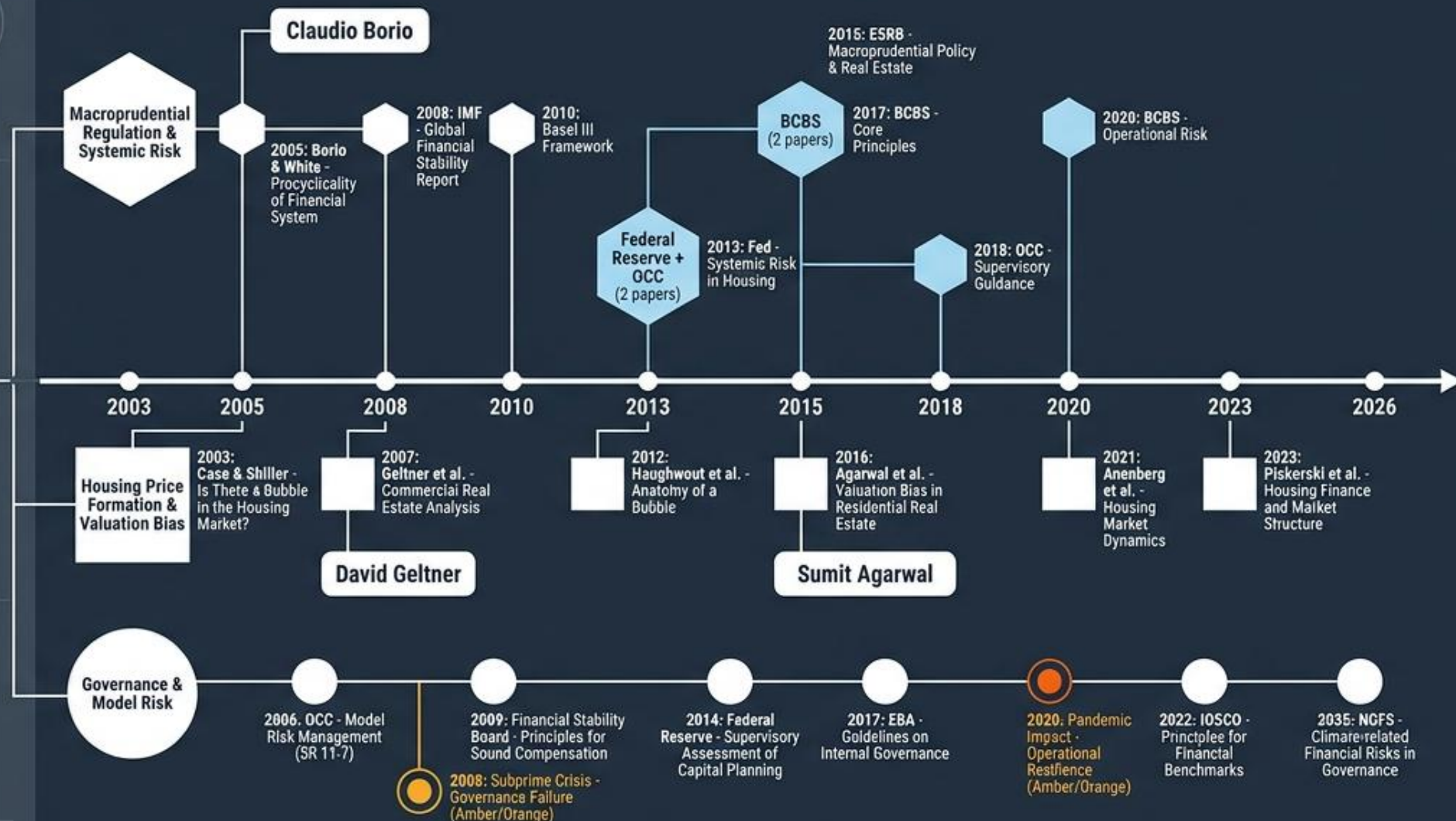
Integrating Valuation Uncertainty into
Credit Risk Analytics via an Institutional
Transmission Mechanism



Designed for Non-Liquid,
Non-Homogeneous Property Markets

Monodeep Saha

The Weight of Evidence



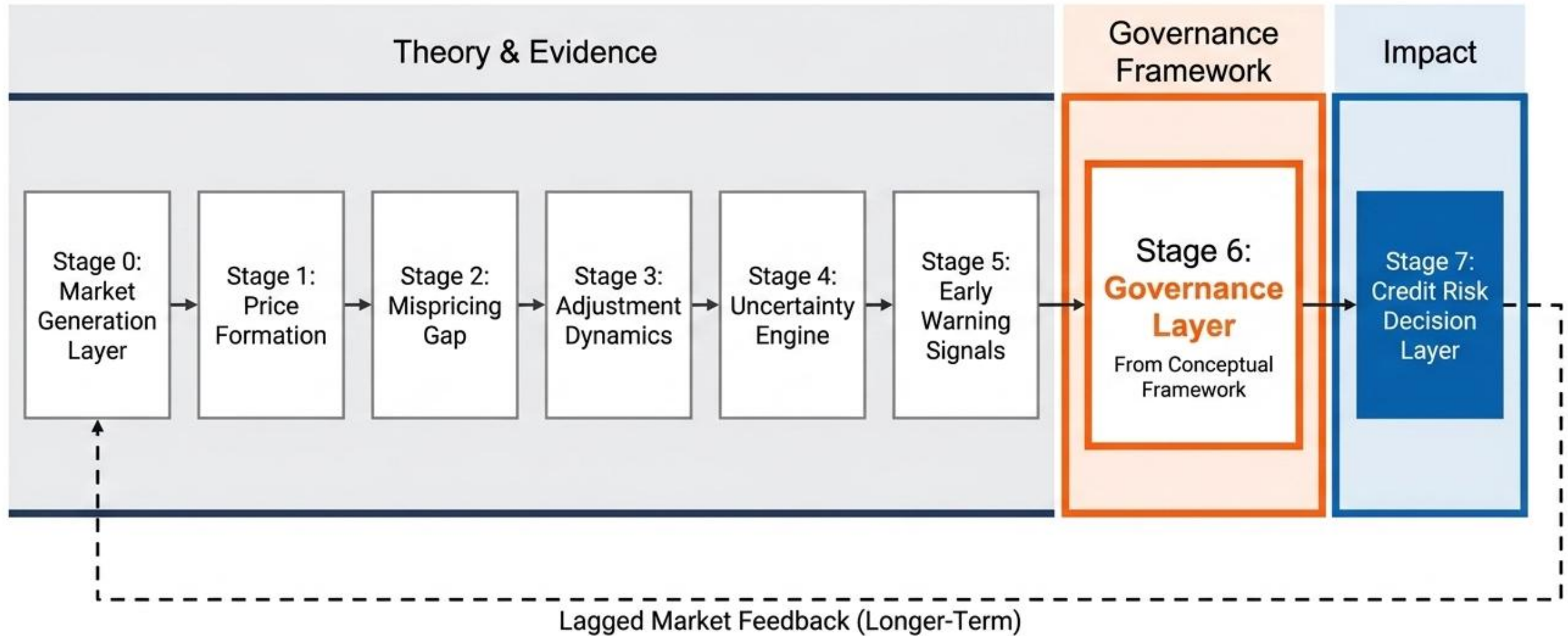
This framework synthesizes 34 foundational studies spanning two decades into a single, actionable governance protocol.

The illusion of precision in non-homogeneous property markets

	Data-Rich / Liquid Markets	Data-Constrained / Illiquid Markets
Price Formation	Fundamental-driven, homogeneous, high transaction volume.	Friction-heavy, heterogeneous, sparse comparable data. 
Mispricing Persistence	Transient, quickly corrected by arbitrage.	Persistent, driven by illiquidity and structural rigidities. 
Valuation Uncertainty	Symmetric, low variance, predictable normal distribution.	Regime-dependent, asymmetric , characterized by fat tails. 
AVM Reliability	High point-estimate confidence.	High uncertainty, fragile to data quality flags. 

Raw AVM outputs in illiquid markets generate volatile point estimates that corrupt downstream credit risk analytics. A new intervention layer is required.

The end-to-end institutional valuation architecture



Translating market behavior into empirical propositions

The Market Reality

The Framework Rule

Question (RQ-A):
How do prices form?



Proposition (P1): Price formation is fundamental-driven but weaker in heterogeneous markets.

Maps to Stage 1

Question (RQ-B):
What determines mispricing?



Proposition (P2): Mispricing is larger, more persistent, and driven by frictions.

Maps to Stage 2

Question (RQ-C):
How does mispricing adjust?



Proposition (P3): Adjustment is slower, state-dependent, and asymmetric to shocks.

Maps to Stage 3

Question (RQ-D):
What is the role of uncertainty?



Proposition (P4): Valuation uncertainty is characterized by fat tails and regime-dependence.

Maps to Stage 4

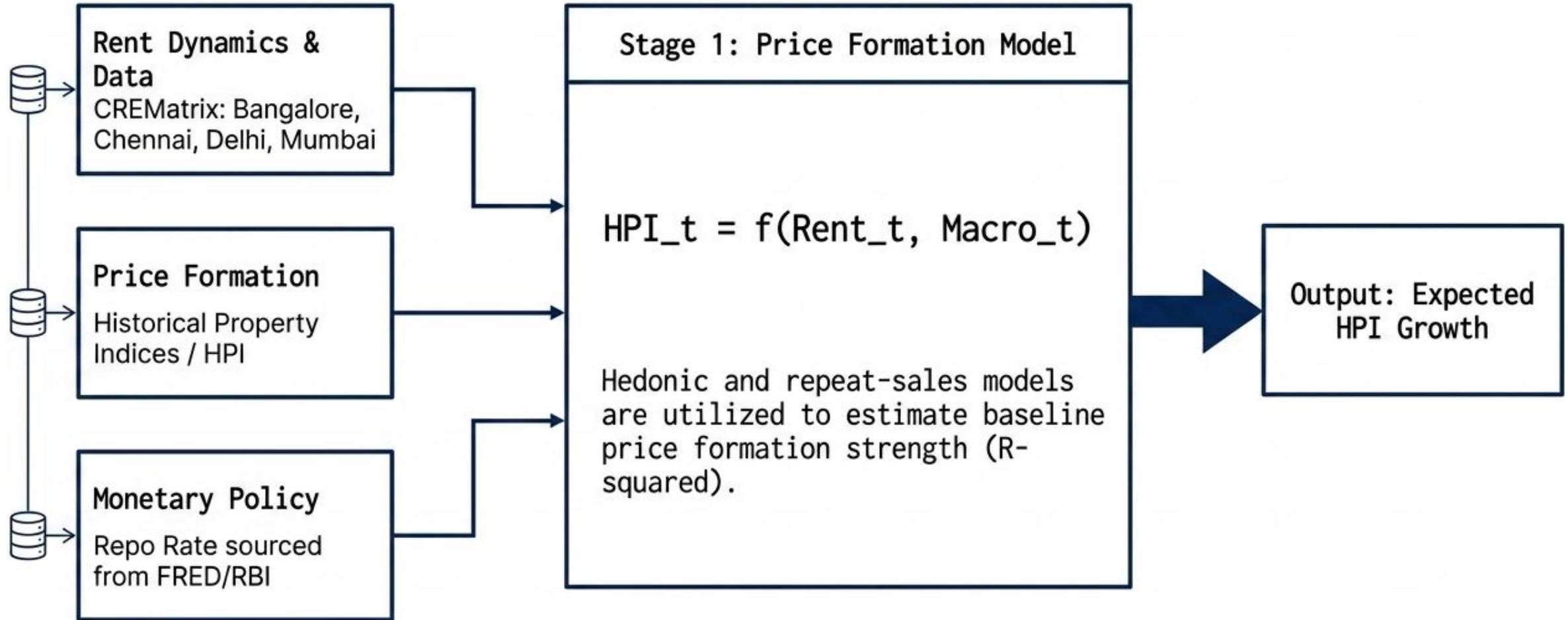
Question (RQ-E):
Can signals predict adjustments?



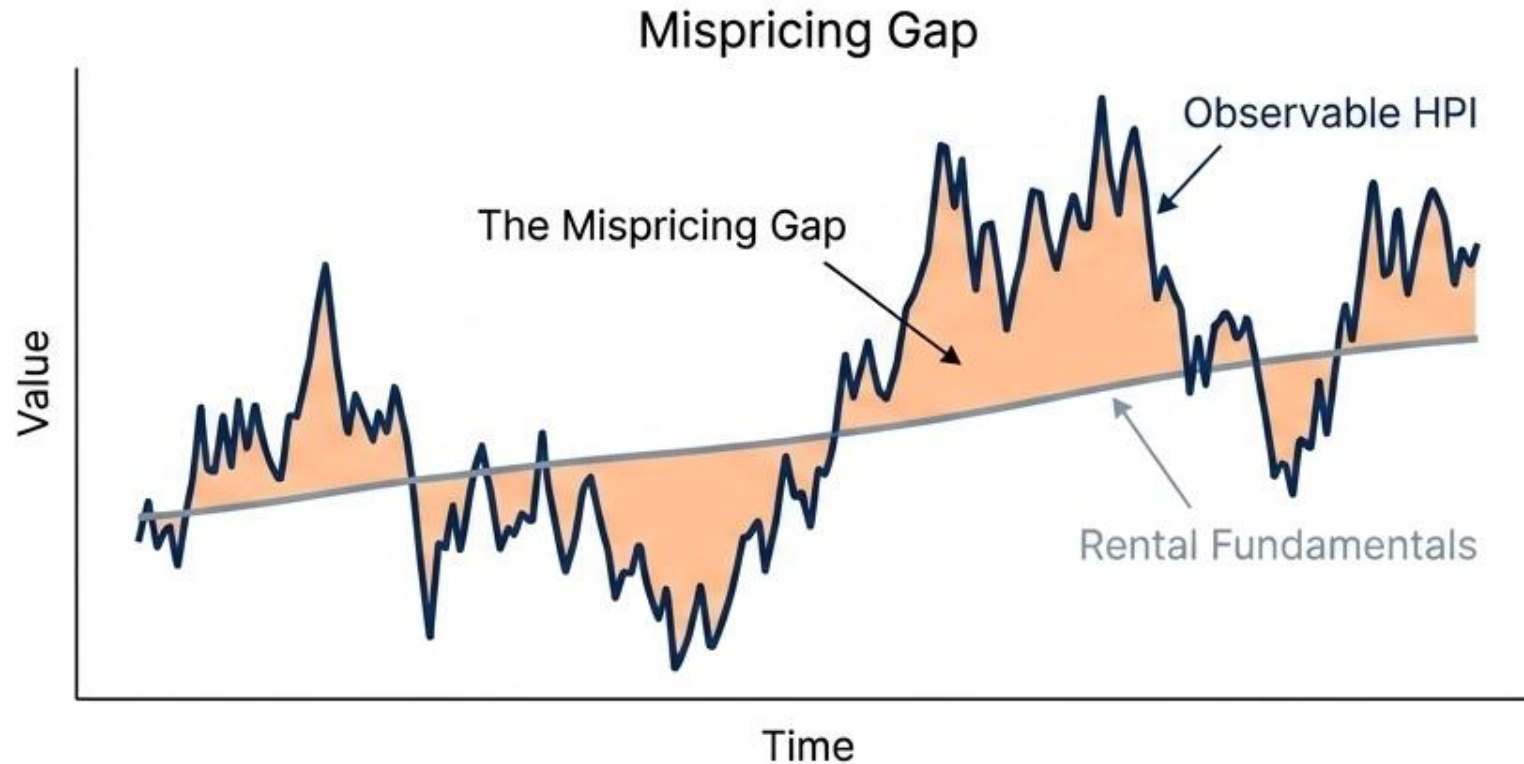
Proposition (P5): Market-based signals provide early-warning indicators.

Maps to Stage 5

Establishing the market baseline and expected growth



Quantifying the deviation between price and fundamentals



$$\text{Gap}_t = \text{HPI}_t - \text{Rent}_t$$

Purpose

To capture and measure the precise deviation between actual price and underlying rental fundamentals.

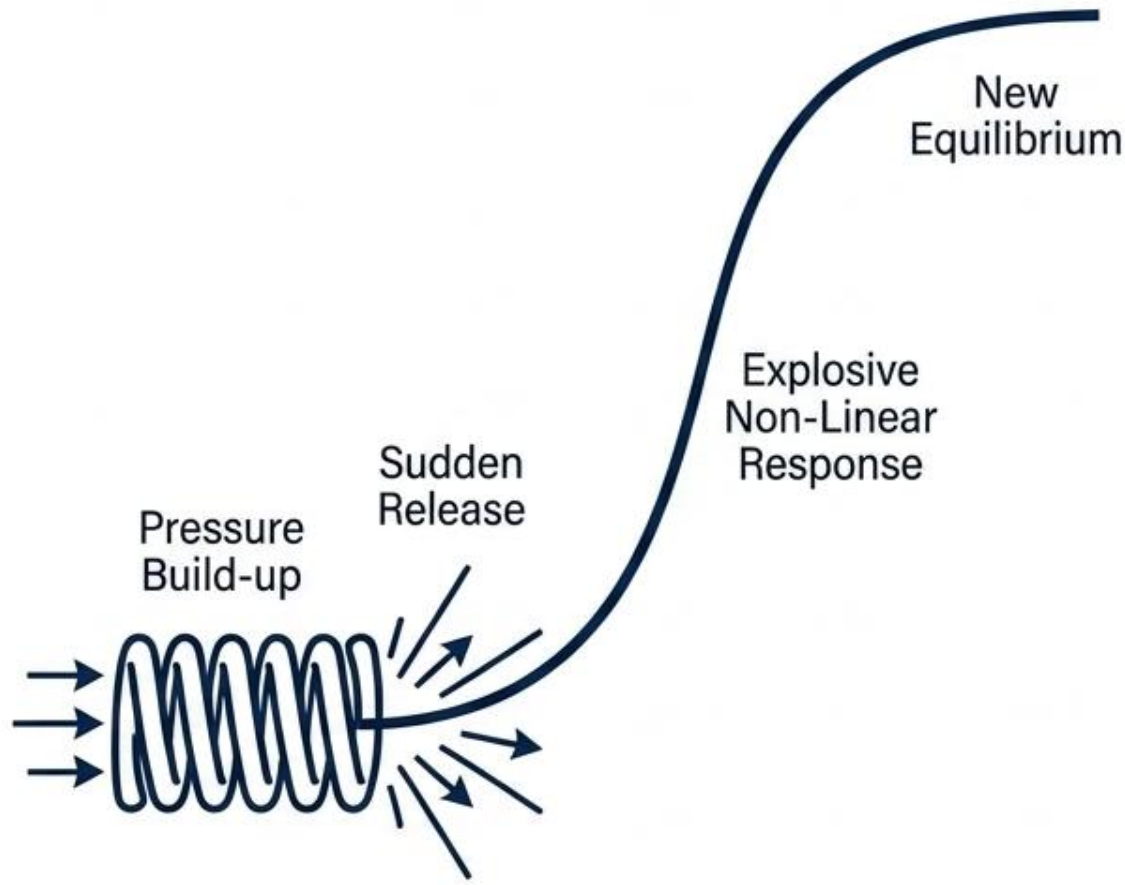
Output

The Valuation Gap becomes the core mathematical state variable for the rest of the model.

Context

In illiquid markets, this gap persists longer and stretches further before correction occurs.

Modeling dynamic adjustments and error-correction



Difference Model:

$$\Delta \text{Gap}_t = f(\Delta \text{Repo}_t, \text{Regime}_t, \varepsilon_{t-1})$$

1 Error-Correction Mechanism

The gravitational mechanism by which the market attempts to pull prices back toward baseline fundamentals after experiencing a shock (ε_{t-1}).

2 Non-Linear Asymmetry

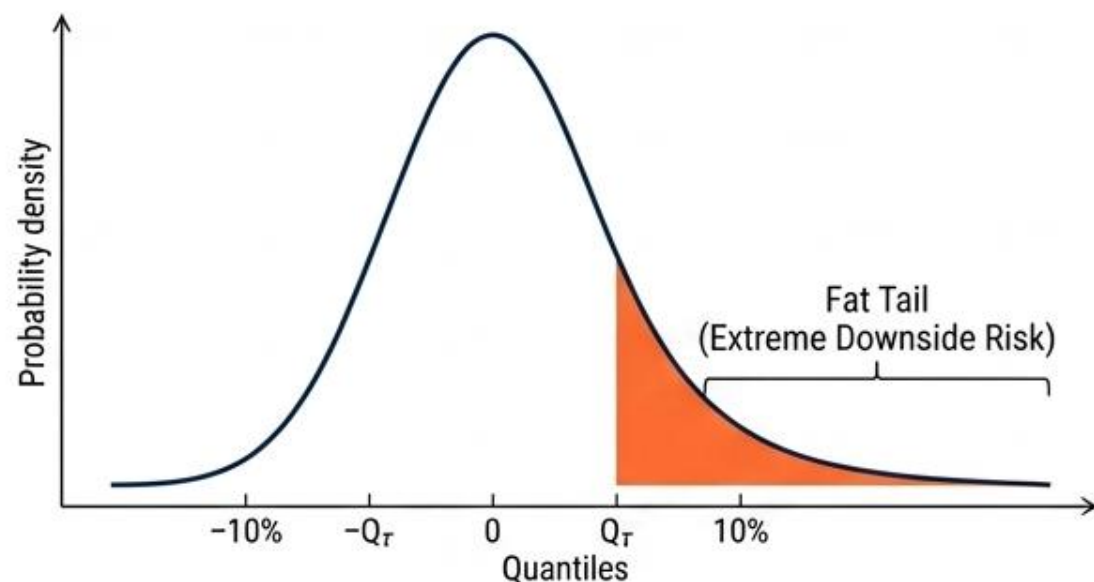
Adjustments are not smooth. They are state-dependent. Markets react violently to sudden shocks (e.g., rate hikes) while recovering sluggishly during gradual easing.

Output: The model dictates the speed, symmetry, and magnitude of adjustment, generating the ΔGap signal.

Appendix

The Uncertainty Engine: Capturing tail risk and volatility

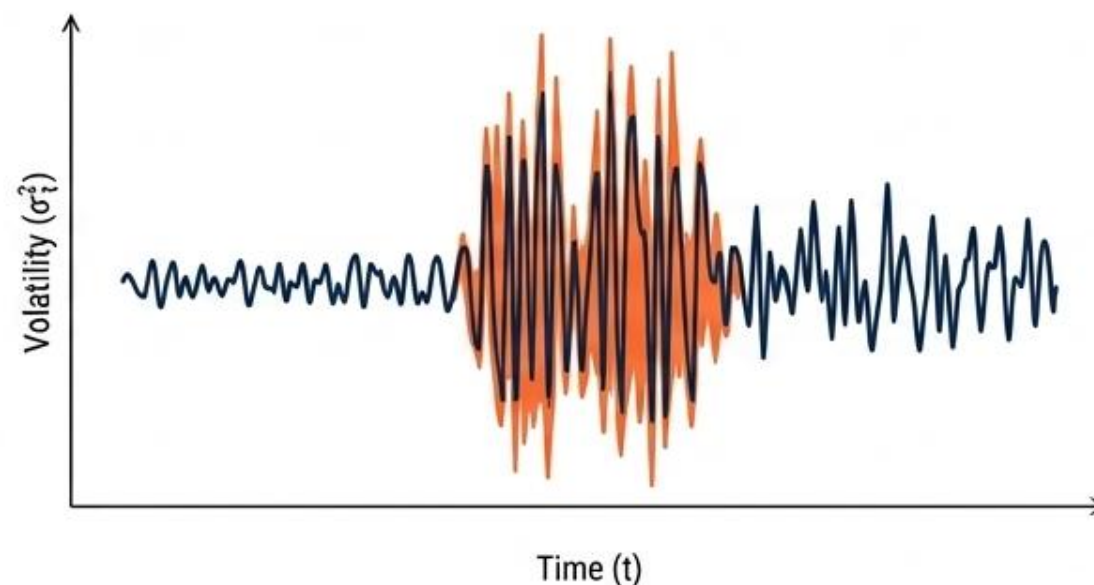
Distributional Model (Quantile Regression)



$$Q_\tau(\Delta Gap_t)$$

Models tail behavior, skewness, and extreme dispersion across different quantiles. Crucial for measuring asymmetric downside valuation risks.

Volatility Model (GARCH(1,1))

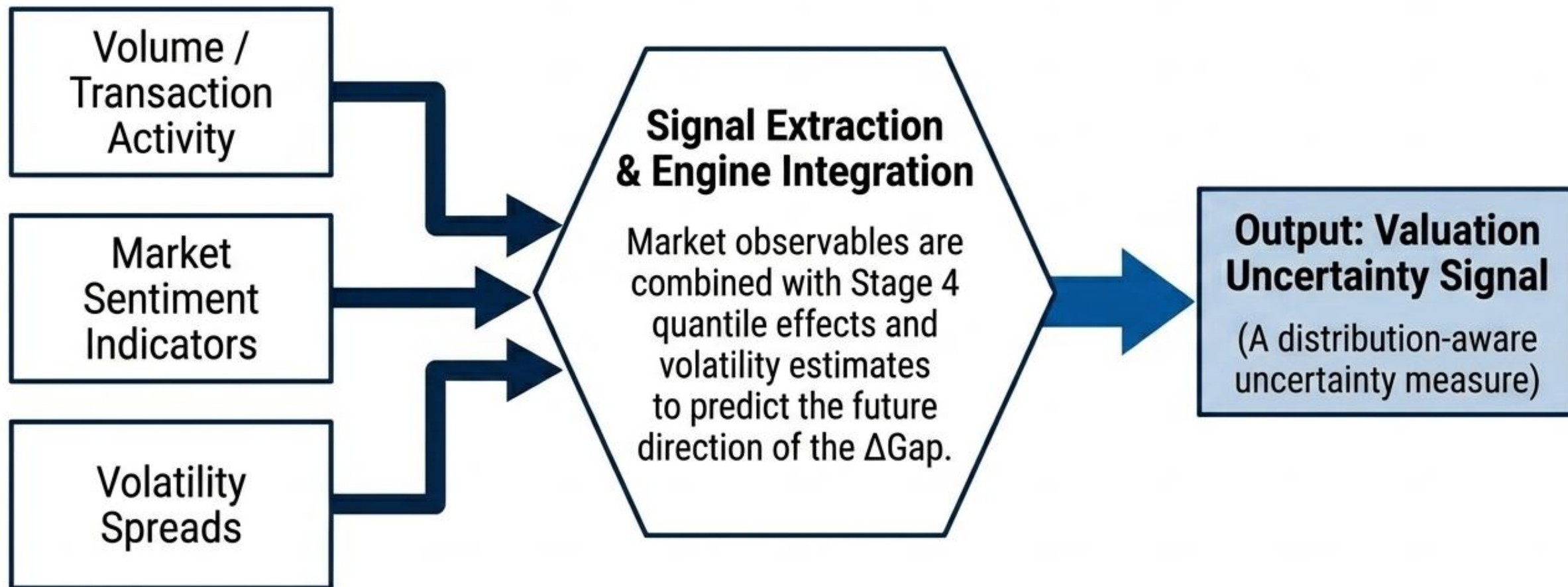


$$\sigma_t^2 = f(\epsilon_{t-1}^2, \sigma_{t-1}^2, \Delta Repo_t, Regime_t)$$

Captures time-varying uncertainty, regime-dependent sensitivity, and the persistence of valuation shocks over time.

Combined Output: These concurrent models generate distribution-aware, volatility-based uncertainty measures.

Extracting valuation signals and early warnings

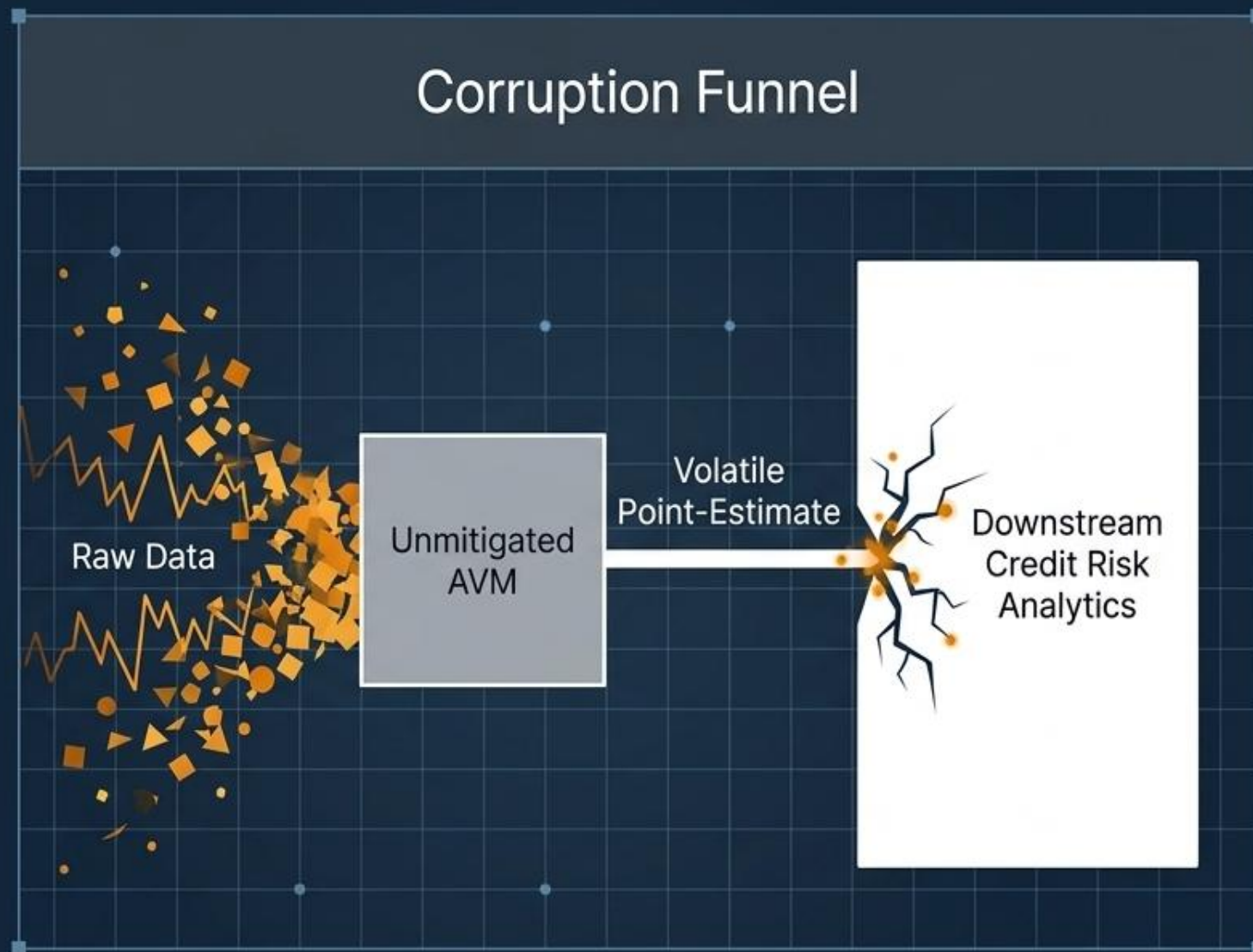


- **Key Insight: Early signals in heterogeneous markets** possess **significant predictive power**, categorizing the valuation as Ready to be Accepted, Adjusted, or Escalated.

The Entropy Problem

Raw Market Uncertainty

In data-constrained, illiquid property markets, precision is absent. Raw data inputs lack the density and homogeneity required for some community required for stable algorithmic processing.



Raw AVM outputs in illiquid markets generate volatile point estimates that corrupt downstream credit risk models.

The Architectural Shift

Valuation uncertainty is not a statistical bug to be ignored; it is a structural feature that must be actively managed.

To protect credit parameters, institutions must construct a filtration mechanism between raw market data and the final risk models.



Moving from chaotic point-estimates to governed signals requires a rigid, four-pillar institutional framework.

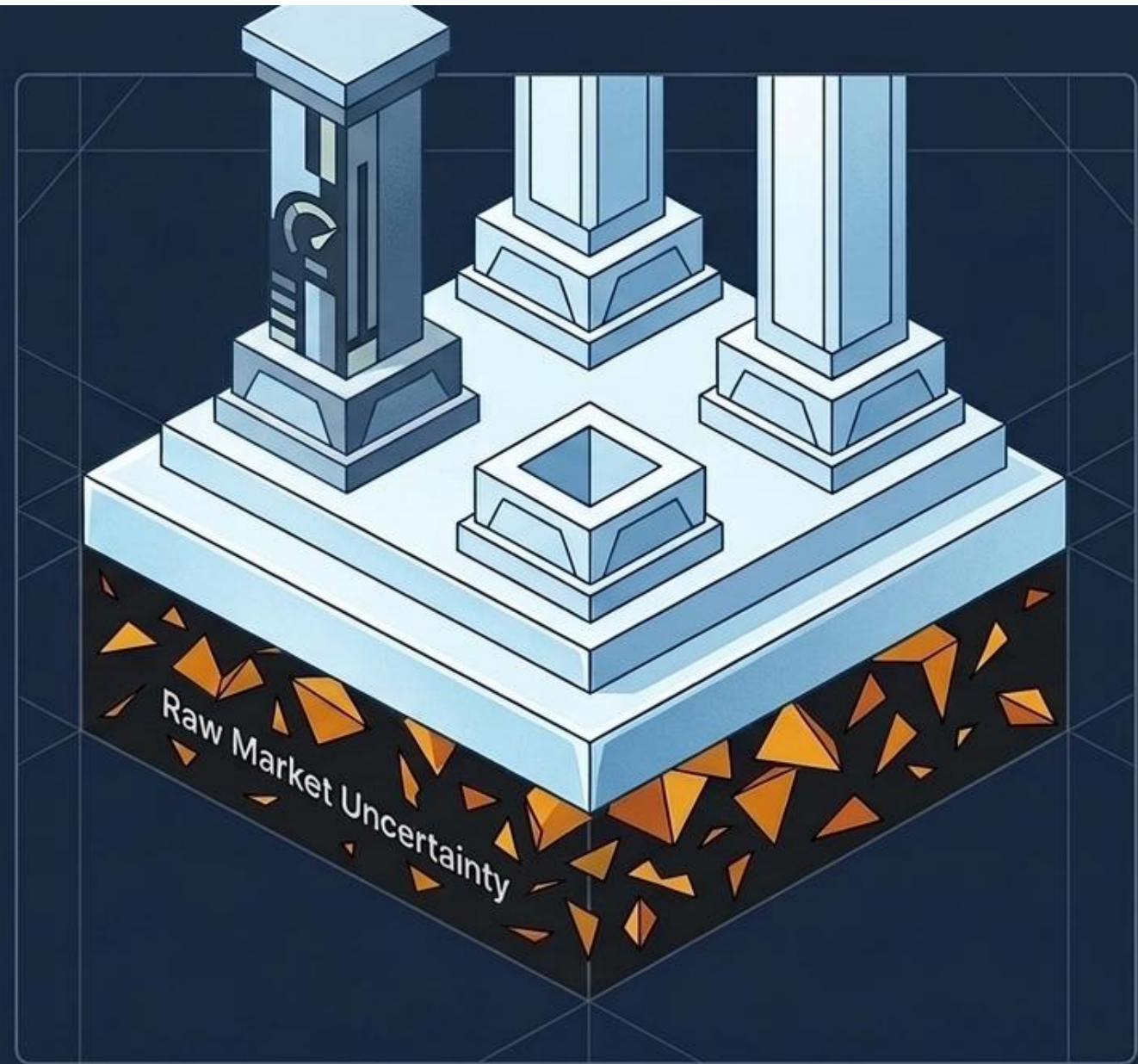
Pillar I: Override Policies & Risk Thresholds

Mechanism:

Establishes strict rules for human judgment and quantitative relevance bands for model uncertainty.

Impact:

Prevents automated models from operating blindly outside of their trained data constraints.



Governance requires hard quantitative boundaries where algorithmic autonomy ceases and human oversight begins.

Pillar II: Liquidity Haircuts

Mechanism:

Applies specific downward adjustments to account for market illiquidity and transaction time frictions.

Impact:

Directly counters the persistent mispricing identified in illiquid property markets, ensuring collateral values reflect realizable distressed pricing.



Illiquidity is a quantifiable friction cost that must be systematically deducted from raw valuation estimates.

Pillar III: Expert Review & Adjustments

Mechanism:

Enforces strict protocols for human escalation, model risk overrides, and rigorous documentation.

Impact:

Provides a documented, auditable trail for when models breach risk thresholds and require human intervention to establish the final value.



Expert human judgment is not arbitrary; it must be channeled through strict, documented escalation protocols.

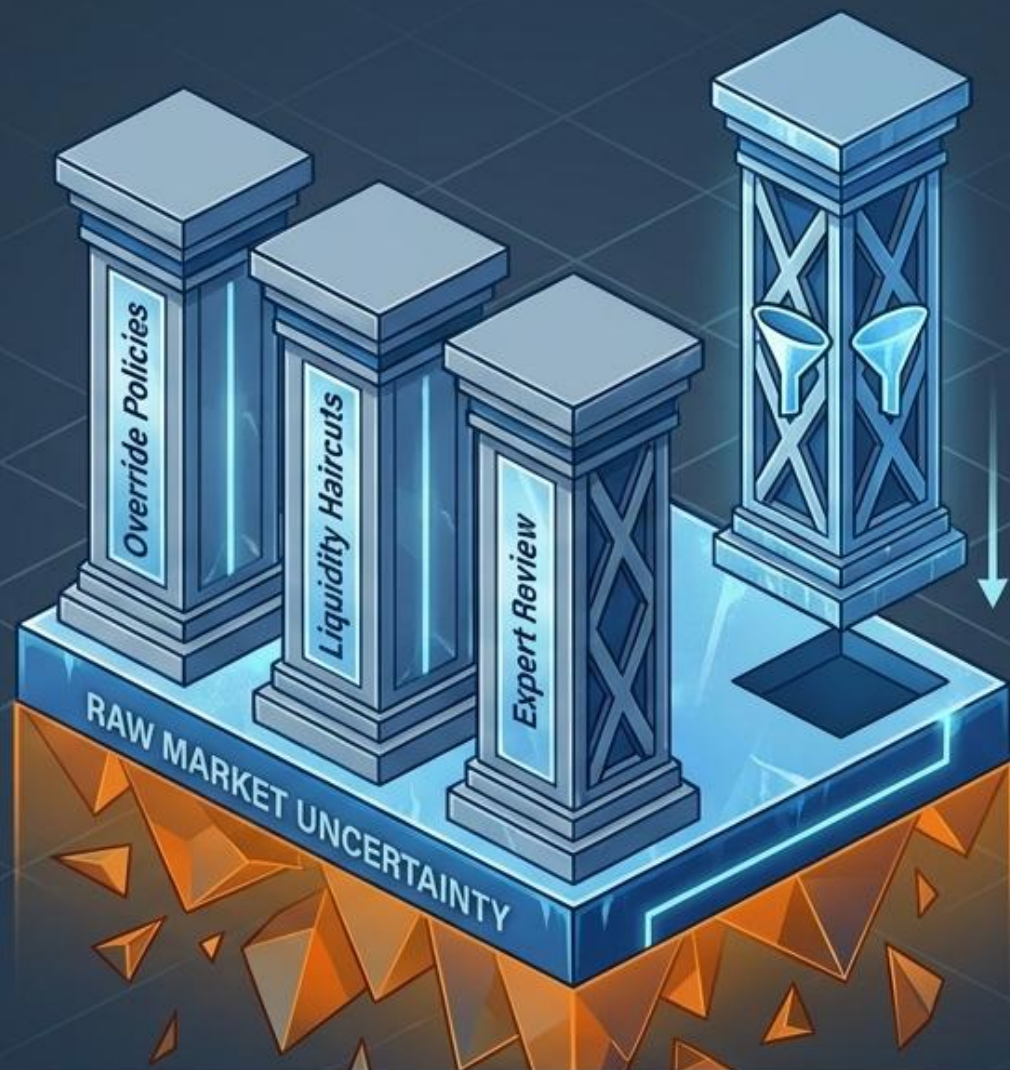
Pillar IV: Mandatory Institutional Translator

Mechanism:

Converts inherent market reality into robust, risk-sensitive credit decisions and capital allocation.

Impact:

Ensures that the adjustments made in the previous three pillars are standardized into a format that core banking systems can reliably ingest.



A governed valuation is useless unless structurally translated into a standard institutional format.

The Governed Valuation Signal

The output is no longer a volatile point estimate. It is a distribution-aware uncertainty measure that is stable, consistent, and ready for credit ingestion.

Entropy has been successfully converted into governed structural order.



The governance framework transforms fragile point estimates into distribution-aware, stable signals.

Integration into Credit Risk Analytics and Decisions



Collateral Haircuts & LTV Calibration

Governed valuation signals dictate dynamic Loan-to-Value limits and apply strict collateral discounting algorithms based on the current market uncertainty regime.



LGD Assumptions

Loss Given Default parameters are continuously updated to reflect the true, friction-adjusted realization value of the underlying collateral property.



ECL Estimation (IFRS 9)

Governed uncertainty inputs directly feed into Expected Credit Loss models, ensuring absolute compliance with forward-looking regulatory accounting requirements.



Pricing, Limits, & Capital Allocation

Final approved valuation signals and expert judgment overlays inform the ultimate credit decision, categorizing applications as Accepted, Adjusted, Declined, or Monitor.



Accepted



Adjusted

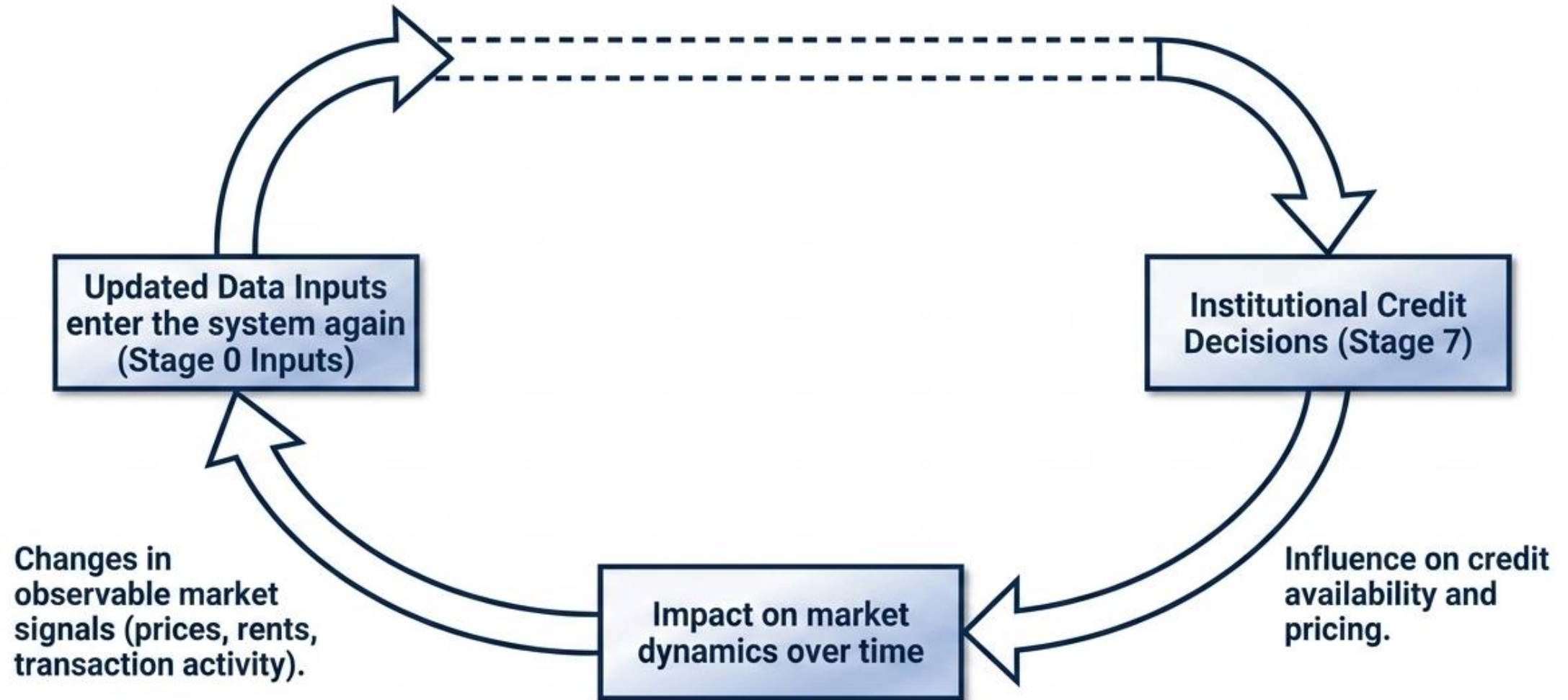


Declined



Monitor

The longer-term market feedback loop



Note: This feedback operates over a longer horizon through macro market conditions. It does not alter governance decisions contemporaneously.

A new paradigm for institutional risk architecture

The Reality

Illiquid, non-homogeneous property markets generate inherent, persistent valuation uncertainty that standard AVMs simply cannot resolve.



The Mechanism

Robust quantitative engines (GARCH, Quantile Regression) can successfully measure and map this uncertainty, but they cannot independently make credit decisions.



The Mandate

Valuation Governance is not an optional compliance exercise; it is the mandatory operational transmission layer required to translate market uncertainty into robust, IFRS 9-compliant credit risk decisions.



Q n A

The End